Data and Applications

Project Phase 1

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Abstract

This document outlines the requirement analysis for a database designed to manage information related to ghouls and their interactions with humans, particularly focusing on the needs of the Commission of Counter Ghoul (CCG) (based on Tokyo Ghoul). The database aims to track various entities such as ghouls, humans, investigators, wards, and their relationships. It also includes functional requirements for data retrieval, analysis reports, and modification operations to ensure efficient management of ghoul-related activities.

Introduction

This database keeps track of various interactions in the world of "Tokyo Ghoul", a show which based on the co-existence of humans and human-eating ghouls – who look almost indistinguishable from humans. The Commission of Counter Ghoul (CCG) is a (human) federal agency that aims to investigate ghoul crimes and eliminate ghouls from the country (Japan).

The scope of the mini-world includes all the entities and relationships described below. It does not cover all aspects of life and survival in the fantasy world to reduce complexity. In addition, this mini-world is only based on the universe of "Tokyo Ghoul", not any of the spin-offs like "Tokyo Ghoul: RE".

Purpose of the Mini-world

The main purpose of this mini-world is to model the complex relationships between humans, investigators, ghouls, and other various aspects within the universe. It provides a structured way to store and retrieve information about their characteristics and interactions. The database serves as a key tool for the CCG to monitor and manage ghoul-related activities like basic data about investigators, data and rating (strength) of ghouls, data about their weapons, data about ghoul crimes, data about ghoul encounters with CCG investigators, etc.

Users of the database

The database serves the following users:

1. **The CCG:** CCG investigators can use this database to track the various ghouls they have seen. Since every ghoul has a different weapon (Kagune) with differing strengths (rating), it needs to all be tracked in an efficient way. To counter ghouls, specialised weapons (Quinque) are required, and they are made from Kagune of dead ghouls and can be upgraded multiple times, so that has to be tracked too. Other data, like ghoul encounters, murders, etc. help the CCG track the whereabouts of ghouls to aid them in eliminating them.
2. **Demographic Analysts:** Demographic analysts can use this database since it maps the places where humans and ghouls live.

3. **Researchers:** Researchers working on ghoul behaviour can use this database to study their behaviour like eating habits, fighting habits, etc.
4. **The Government:** The Government can use the data in this database to issue alerts like evacuation notices when a place sees an uptick in ghoul activity.

Purpose of the database

The purpose of the database is to provide a structured and efficient way to store, manage, and retrieve information about ghouls, humans, investigators, wards and their relationships in the Tokyo Ghoul universe. It aims to facilitate tracking of ghoul activities, human demographics, and investigator roles, thereby aiding in research, analysis and operational decision-making. For demographics, it focuses on wards, humans, ghouls, and their hybrids (one-eyed ghouls). It tracks their attributes and relationships to understand their distribution and interactions within the city. For ghoul investigations, it focuses on investigators, their weapons (quinque), ghouls, and encounters. It tracks investigator details, quinque attributes, ghoul characteristics, and encounter outcomes. This helps the CCG monitor ghoul threats, manage investigator resources, and analyze encounter effectiveness. The database is designed to support various users including the CCG, demographic analysts, researchers, and the government, providing them with accurate and up-to-date information to aid in their respective roles. We can easily update/modify the database as new investigators join the CCG, new quinques are made, crime occurs, ratings/strength get updated. This ensures that the database remains relevant and useful over time.

Database Requirements

Strong Entity Types

Ghoul

Ghouls are a carnivorous and cannibalistic humanoid species that are only able to feed on the flesh of humans and other ghouls. They share similarities with humans in physical appearance and intelligence, but have major differences in their inner biology, mentality, and diet.

Ghouls have a strength *Rating* depending on their RC level and kagune. There are six rating levels, from SSS (the most powerful) to C (the weakest), with `+` and `-` subratings for each (e.g. `A+`, `SS-`, etc.). Sometimes, ratings are prefixed with a ~ to indicate that the ghoul is at least that rating, but the exact rating is unknown. All these possible rating values, as indicated by the below regex, are stored in an enumeration called `rating`. This enumeration is created in such a way that ghouls are sortable by rating, i.e. by making sure that the underlying number in the enum for higher ratings are higher.

```
rating = /~?((A|B|S{1,2})(+|-)?|C|SSS)/
```

Attributes:

(Unless otherwise specified, all attributes of all entities in this document are not nullable)

Column	Type	Additional Notes
`id`	`int`	`key attribute`
`name`	`text`	
`birth_date`	`date`	
`age`	`int`	Derived from `birth_date`.
`gender`	`text`	
`deceased`	`boolean`	Whether the ghoul is dead or not.

Column	Type	Additional Notes
`rc_level`	`int`	The ghoul's RC level. A healthy ghoul's will be around 1000-8000, but this isn't a constraint.
`rating`	`rating[]`	The ghoul's rating (strength). Some ghouls have multiple ratings depending on their form.

Kagune

A kagune is a ghoul's predatory organ and functions as their weapon and claws. It is usually as red as blood, flexible like the flow of water, but firm and sturdy. When released, a ghoul's physique is strengthened, they are more resilient, and their mobility heightens.

The kagune's appearance and place of emergence depends on the *RC Type* of the ghoul. there are four possible rc types: `ukaku`, `koukaku`, `rinkaku`, and `bikaku` (these are stored in an enum called `rc_type`). A ghoul is usually born with only one kagune of one RC type, but some ghouls can have multiple kagune of different RC types, for example, ghouls that are born from parents of different RC types (Chimera), and ghouls that canibalise attain special kagune called kakuja. These interactions are all realised through relations described later in the document.

Kakuja are special kagune that are attained by a ghoul when they cannibalize on other ghouls. Kakuja develop and become stronger as ghouls consume more ghouls. This is tracked in the `kakuja_level` of the kagune, which can be `null` if the kagune is not a kakuja.

Attributes:

Column	Type	Additional Notes
`id`	`int`	`key attribute`
`rc`	`rc_type`	
`kakuja_level`	`int`	The kakuja level (see above). It can be null if the kagune is not a kakuja

Human

A Human in this universe, as in any, is a member of society. Humans are the primary food source for ghouls, who can only survive by consuming human flesh. Humans form a superclass to investigators. Humans and Ghouls can breed to produce offspring known as One-Eyed Ghouls.

Attributes:

Column	Type	Additional Notes
`id`	`int`	`key attribute`
`name`	`text`	
`birth_date`	`date`	
`age`	`int`	Derived from `birth_date`.
`gender`	`text`	
`deceased`	`boolean`	Whether the ghoul is dead or not.
`rc_level`	`int`	The person's RC level. A healthy human's will be around 100-500, but this isn't a constraint.

Ward

A ward are subdivisions of cities, modeled after real-life wards, which are subdivisions of Japanese cities. The city of Tokyo has 24 wards in the Tokyo Ghoul universe. Wards in Tokyo Ghoul mostly coincide with real-life Japanese wards. Ghouls usually live in wards they mark as hunting grounds, and there are conflicts between ghouls of different gangs over hunting grounds. Wards are numbered within a city, and they are also given names. We also give wards a unique ID that identify them irrespective of the city.

Attributes:

Column	Type	Additional Notes
`id`	`int`	`key attribute`
`division`	`(number int, city text)`	`key attribute`, the division (e.g. 1st ward of Tokyo)
`name`	`text`	
`boundary`	`polygon`	The ward's boundary, stored using GIS

Investigator

A ghoul investigator is a specially trained professional employed by the CCG. They are tasked with the investigation, pacification, and elimination of ghouls to protect the civilians of Japan from any potential harm caused by ghouls. During their investigations, they analyze crime scenes for traces of ghoul perpetrators, expose their identities, incapacitate or neutralize them. Investigators are a **subclass** of humans, so an investigator is a human.

Investigators can be junior or senior, and like how ghouls have a rating, investigators have a "rank" that indicates how strong they are, and the responsibilities they have. Investigator ranks are in three levels, per seniority, and rank names depend on seniority.

Rank	Senior Rank Name	Junior Rank Name
`1`	`Special Class`	`Rank 1`
`2`	`Associate Special Class`	`Rank 2`
`3`	`First Class`	`Rank 3`

Rank Strength hierarchy goes as follows: `Rank 3` < `Rank 2` < `Rank 1` < `First Class` < `Associate Special Class` < `Special Class`.

Attributes:

Column	Type	Additional Notes
`division`	`(number int, city text)`	Their primary division (e.g. 1st ward of Tokyo)
`is_senior`	`boolean`	
`rank`	`int`	Allowed values: [1, 2, 3]

Quinque

Ghouls cannot be harmed using traditional human weapons. A ghoul can only be harmed using kagune, or weapons made using kagune. A quinque is such a weapon used by CCG ghoul investigators. They can be made into some type of melee or ranged weapon, but most retain a few characteristics from the original kagune. To attain a quinque, an investigator must first confirm their target's Rc type before exterminating them. Afterwards, it is common to name the quinque from the ghoul's name or alias. Like ghouls, quinque also have a rating. Quinque can be upgraded by using kagune of other ghouls.

Column	Type	Additional Notes
`id`	`int`	`key attribute`
`name`	`text`	
`rating`	`rating`	

Report

These are daily reports published by the CCG regarding their actions performed on a day.

Column	Type	Additional Notes
`id`	`int`	`key attribute`
`date`	`date`	`key attribute`
`text`	`text`	The text of the report

Weak Entity Types

OneEyedGhoul

One-eyed ghouls are ghoul-human hybrids either born as a child to a human and a ghoul or artificially formed via transplant surgery. They act like other ghouls, including possession of a kagune, but they also exhibit some human qualities, like being able to consume human food. This human-ghoul hybrid will have a human counterpart in the `Humans` table, and a ghoul counterpart in the `Ghouls` table. These two entities will be the identifying relationships of this entity. In addition to these, it also relates to the transplant donor ghoul, if there was a transplant, or it relates to the two parents of the ghoul.

This entity is an entity and not a relationship because one-eyed ghouls are a special class of ghouls with their own unique attributes such as having one kakugan (red eye) instead of two (hence the name).

Relationships:

1. **GhoulPart** (identifying): The ghoul counterpart of this one-eye.

Degree	`2`	
Participants	`OneEyedGhoul`	`Ghoul`
Cardinality Ratio	`1`	`1`
Participation Constraint	`Total`	`Total`

2. **HumanPart** (identifying): The human counterpart of this one-eye.

Degree	`2`	
Participants	`OneEyedGhoul`	`Human`
Cardinality Ratio	`1`	`1`
Participation Constraint	`Total`	`Total`

3. **TransplantDonor**: If this one-eye was formed as a result of a transplant, this is the transplant donor ghoul. The transplanted organ is a ghoul's kakuho, the organ responsible for generating the kagune. Some ghouls may have multiple kakuho, and thus, can donate multiple times, but to different ghouls, hence it's a `1:N` ratio.

Degree	`2`	
Participants	`OneEyedGhoul`	`Ghoul`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Partial`	`Partial`

4. **GhoulParent**: If this one-eye was formed as a result of a child between a ghoul and a human, this is the ghoul parent. A ghoul can have multiple one-eyed children, but a one-eye has only one ghoul parent, so `1:N`.

Degree	`2`	
Participants	`OneEyedGhoul`	`Ghoul`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Partial`	`Partial`

5. **HumanParent**: If this one-eye was formed as a result of a child between a human and a ghoul, this is the human parent. A human can have multiple one-eyed children, but a one-eye has only one human parent, so `1:N`.

Degree	`2`	
Participants	`OneEyedGhoul`	`Human`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Partial`	`Partial`

While `TransplantDonor`, `HumanParent`, and `GhoulParent` are all marked as `Partials`, one-eyes have to be formed by either method, so either `TransplantDonor` XOR (`HumanParent` and `GhoulParent`) should be set for a one-eyed ghoul.

GhoulMurder

This entity tracks all the (reported) murders committed by ghouls. Humans comprise a majority of a ghoul's diet. Although, a ghoul may kill a human for other reasons as well. It is a weak entity with identifiying relationships with the ghoul and human in question.

Attributes:

Column	**Type**	**Additional Notes**
`time`	`datetime`	Time of Murder

Relationships:

1. **Murderer** (identifying): The ghoul who murdered.

Degree	`2`	
Participants	`GhoulMurder`	`Ghoul`
Cardinality Ratio	`N`	`1`
Participation Constraint	`Total`	`Partial`

2. **Victim** (identifying): The human who was murdered.

Degree	`2`	
Participants	`GhoulMurder`	`Human`
Cardinality Ratio	`N`	`1`

Participation Constraint	`Total`	`Partial`

3. **Ward**: The ward in which this murder was committed in.

Degree	`2`	
Participants	`GhoulMurder`	`Ward`
Cardinality Ratio	`N`	`M`
Participation Constraint	`Total`	`Partial`

Encounter

Investigators target ghouls in teams of two – a senior and a junior. This entity tracks the various fights between investigators and ghouls.

Attributes:

Column	Type	Additional Notes
`time`	`datetime`	Time of Encounter
`death_senior`	`boolean`	Whether the battle resulted in the death of the senior
`death_junior`	`boolean`	Whether the battle resulted in the death of the junior
`death_ghoul`	`boolean`	Whether the battle resulted in the death of the ghoul

Relationships:

1. **Senior** (identifying): The senior investigator in the battle.

Degree	`2`	
Participants	`Encounter`	`Investigator`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Total`	`Partial`

2. **SeniorQuinque**: The quinque the senior investigator used.

Degree	`2`	
Participants	`Encounter`	`Quinque`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Total`	`Partial`

3. **Junior** (identifying): The junior investigator in the battle.

Degree	`2`	
Participants	`Encounter`	`Investigator`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Total`	`Partial`

4. **JuniorQuinque**: The quinque the junior investigator used.

Degree	`2`	
Participants	`Encounter`	`Quinque`

Cardinality Ratio	`1`	`N`
Participation Constraint	`Total`	`Partial`

5. **Ghoul** (identifying): The ghoul that was fought.

Degree	`2`	
Participants	`Encounter`	`Ghoul`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Total`	`Partial`

Relationships

Ghoul-Has-Kagune

A ghoul may have more than one kagune (in the case of kakuja and chimera kagune), but a kagune can only belong to one ghoul, thus the cardinality ratio is `1:N`. As all ghouls necessarily have a kagune, the relationship has total participation.

Degree	`2`	
Participants	`Ghoul`	`Kagune`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Total`	`Total`

Ghoul-LivesIn-Ward

Ghouls usually confine themselves to wards they mark as hunting grounds. This is a `N:1` relationship since a ghoul confines themselves to one ward, while a ward can have multiple ghouls. This relationship has partial participation on both ends since sometimes, the wards of ghouls is unknown, and some wards may not have any ghouls.

Degree	`2`	
Participants	`Ghoul`	`Ward`
Cardinality Ratio	`N`	`1`
Participation Constraint	`Partial`	`Partial`

Quinque-MadeFrom-Kagune

A quinque is a weapon which is made from kagune of a ghoul, which is used by investigators to fight other ghouls. A kagune can only make one quinque, and a quinque can only be made from one kagune (though it can be upgraded using other kagune, which is covered later), hence it's a `1:1` relationship. A quinque requires a kagune to be made, while not all kagune are made into quinque.

Degree	`2`	
Participants	`Quinque`	`Kagune`
Cardinality Ratio	`1`	`1`
Participation Constraint	`Total`	`Partial`

Investigator-owns-Quinque

Investigators use quinque as weapons. An investigator can own multiple quinque, but a quinque is owned by exactly one investigator. Not all investigators necessarily own quinque, and not all quinque are necessarily owned by an investigator (e.g. if it is decommissioned).

Degree	`2`	
Participants	`Investigator`	`Quinque`
Cardinality Ratio	`1`	`N`
Participation Constraint	`Partial`	`Partial`

Chimera

Some ghouls have chimera kagune, which happen in extremely rare cases that occur when a ghoul has parents that possess different RC types and inherits both. The offspring has kagune of different RC types and can use them simultaneously. This relation has five participants: the ghoul, the two kagune, and the ghoul's two parents.

Degree	`5`				
Participants	`Ghoul`	`Kagune`	`Kagune`	`Ghoul`	`Ghoul`
Cardinality Ratio	`1`	`1`	`1`	`N`	`N`
Participation Constraint	`Partial`	`Partial`	`Partial`	`Partial`	`Partial`

Killing

This relationship describes a killing of a ghoul by an investigator in a ward using a quinque. This is a 4-degree relationship with cardinality ratio of `1:N:M:P` (since a ghoul can only be killed once) with partial participation on all four participants.

Degree	`4`			
Participants	`Ghoul`	`Investigator`	`Ward`	`Quinque`
Cardinality Ratio	`1`	`N`	`M`	`P`
Participation Constraint	`Partial`	`Partial`	`Partial`	`Partial`

QuinqueUpgrades

A quinque can be upgraded using kagune of other ghouls. Each upgrade increases the quinque's strength. It is a 3-degree relationship over the quinque upgraded, the kagune used to upgrade it, and the investigator who performed (or supervised) the upgrade. A kagune may only upgrade one quinque, while a quinque may be upgraded many times, and an investigator can oversee many upgrades.

Attributes:

Column	Type	Additional Notes
`time`	`datetime`	Time of upgrade

Relationship:

Degree	`3`		
Participants	`Quinque`	`Kagune`	`Investigator`
Cardinality Ratio	`N`	`1`	`M`
Participation Constraint	`Partial`	`Partial`	`Partial`

Note: Other relationships that were already mentioned above (in the weak-entities section) are not repeated here.

Functional Requirements

Retrieval Operations

Selection

- "Select All Ghouls with Rating >= S": Retrieves all the ghouls with a rating equal or higher than S.
- "Select All Wards in Tokyo": Retrieves all the wards where the division city is Tokyo.
- "Select All Ghouls in Ward 1 in Tokyo": Retrieves all the ghouls who are in Tokyo's 1st ward using the `Ghoul-LivesIn-Ward` relationship.
- "Select All Investigators that are Seniors": Retrieves all investigators with `is_senior` set to `true`.

Projection

- "Display the names and birth dates of all Humans":Lists out all basic information such as names and birth dates of humans.
- "List the names and ratings of all Quinques": Lists out all basic information such as names and ratings of quinques.
- "Display the names and numbers of all Wards not in Tokyo": Lists out the names and numbers of the words based in the city of Tokyo.
- "List the names and RC levels of all Ghouls": Lists out all the names and RC levels of ghouls known at the time.

Aggregation

- "Display the average rating of all Ghouls": Calculates the average rating of all ghouls in the database, can be used in case of a major war for strategizing.
- "Select the Ghoul with highest rating": Retrieves the ghoul with the highest rating, useful for identifying top threats.
- "Select the Quinque with lowest rating": Retrieves the quinque with the lowest rating, useful for identifying weapons that may need upgrades or replacements.
- "Count the number of Ghouls in the 1st Ward of Tokyo": Counts the number of ghouls residing in the 1st ward of Tokyo, useful for future planning and tracking.

Search

- "Find all Humans with names ending in `soku`": Searches for humans whose names end with the substring `soku` may return names like- `Tsukiyama Soku`.
- "Find all Reports with text `killed ghoul`": Searches for reports that contain the phrase `killed ghoul` in their text, may be used to populate successful ventures for book-keeping.
- "Find all wards with names having `iwa`": Searches for wards whose names contain the substring `iwa`, may return names like- `Shin-iwa`.

Analysis Report

Note: The following SQL code is of the PostgreSQL dialect

Report A (Investigators killing report in the past year)

Description: For each investigator, count how many ghouls they have killed, and the average rating of the ghouls they have killed.

Actionable Insights: Allows the agency to reward investigators with more kills, judge investigator skill level and ranking from the ghouls they have killed, and put their quinques in line for upgrades if required.

```
select
  i.id,  -- this is the human id, we assume that human attributes are available in
the investigator table
  i.name,  -- human name
  i.is_senior,
  count(e.*) as ghouls_killed,
  avg(g.rc_level) as ghoul_avg_rc,
  avg(g.rating) as ghoul_avg_rating
from investigators i
left join encounters e on (e.senior = i.id or e.junior = i.id) and e.death_ghoul =
true and e.time >= current_date - interval '12 months'
left join ghouls g on g.id = e.ghoul_id
group by i.id, i.name
order by quinques_acquired desc nulls last;
```

Report B (Monthly Encounter Outcomes by Ward and Investigator Rank)

Description: Show where investigator-pairs are fighting ghouls, how often encounters result in ghoul death vs investigator death, and which investigator ranks are involved — helps allocate teams and quinques by ward.

Actionable Insights: If a ward shows low % ghoul kills and higher investigator deaths with mostly low-ranked seniors/juniors, reassign higher-rank teams or equip different quinques.

```
select
  date_trunc('month', e.time) as month,
  w.id as ward_id,
  w.division as ward_division,
  inv_s.rank as senior_rank,
  inv_j.rank as junior_rank,
  count(*) as total_encounters,
  sum(case when e.death_ghoul then 1 else 0 end) as ghoul_kills,
  sum(case when e.death_senior then 1 else 0 end) as senior_deaths,
  sum(case when e.death_junior then 1 else 0 end) as junior_deaths,
  round(100.0 * sum(case when e.death_ghoul then 1 else 0 end) / coalesce(count(*),
0), 1) as pct_ghoul_kills
from encounters e
join wards w on e.ward = w.id
join investigators inv_s on e.senior = inv_s.id
join investigators inv_j on e.junior = inv_j.id
group by month, w.division, w.city, inv_s.rank, inv_j.rank
order by month desc, w.city, w.division;
```

Report C (Top Offender Ghouls and Ward Hotspots)

Description: Identify which ghouls commit most murders, correlate with ward locations for further analysis and inference.

Actionable Insights: If a ward shows high murders by high-rating ghouls, prioritize high class-rank investigator deployment and quinque upgrades there.

```
 select
  g.id as ghoul_id,
  g.name as ghoul_name,
  g.rating as ghoul_rating,
  w.id as ward_id,
  w.division as ward_division,
  count(gm.*) as murders_reported,
  min(gm.time) as first_reported,
  max(gm.time) as last_reported
from ghoul_murders gm
join ghouls g on gm.ghoul = g.id
join wards w on gm.ward = w.id
group by g.id, g.name, g.rating, w.division, w.city
order by murders_reported desc, g.rating desc
limit 50;
```

Modification Operations

Insertion:

- Insert a new human with their details (id, name, birth_date, gender). No need to add age as it is derived. Check id for uniqueness, it should be of integer data type. Check if the birth_date is a valid date. Check if the name is a string and gender is text.
- Insert a new investigator with their details (division, rank, is_senior) in the investigator table. Check for id for uniqueness. It should be of integer data type. Check if the is_senior attribute is a boolean and rank is an integer between 1-3.
- Insert a new quinque with its details (id, name, rating) in the quinque table. Check for id for uniqueness. It should be of integer data type. Check if the rating is one of the enum values. Check if the name is a string. Check if the referenced ghoul_id in the relationship exists in the ghoul table.
- Insert a ghoul_murder entry with attributes as ghoul_id, human_id, eand time of murder, ensuring the referenced ids exist in the ghoul and human tables.

Update:

- Update an existing investigator's details (division, rank, is_senior) in the investigator table.
- Update rating of a ghoul (their rating can change as they grow stronger, or attain a kakuja).
- Update the rating of a quinque (their rating can change if it was upgraded).
- Update the rc_level of a human.
- Update the rc_level of a ghoul.

Deletion:

- Delete an investigator from the investigator table (reason of deletion could be that they got fired/retired).
- Delete a report if it was accidentally published.